# Homework 1 — changed numbers (extra practice)

Module 1 - Introduction & Math Review · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A pharmacist dissolved 300 milligrams of a new antibiotic drug into a volume of purified water and placed the solution in a refrigerator (4°C). At various time intervals, the pharmacist removed a small aliquot from the solution and measured the amount of drug contained in each aliquot. The following data were obtained.

| Time (hr) | Drug (mcg/mL) |
|---|---|
| 0.5 | 226.6 |
| 1 | 213.9 |
| 2 | 190.7 |
| 4 | 151.5 |
| 8 | 95.6 |
| 12 | 60.4 |
| 16 | 38.1 |

**1a** Is the decomposition of this antibiotic a first-order or a zero-order process?

A. Zero order

B. First order

C. Neither: it needs a third plot

**1b** What is the rate constant k for the decomposition of this antibiotic?

**1c** What is the half-life t½?

**1d** What was the initial volume of water used to prepare the original solution?

**1e** How much time is required for the concentration of drug to decrease by 75%?

2. Below is the decrease in the amount of drug as a function of time.

| Time (hr) | Drug A (mg) |
|---|---|
| 0.5 | 508 |
| 1 | 496 |
| 2 | 472 |
| 3 | 448 |
| 6 | 376 |
| 9 | 304 |
| 12 | 232 |
| 16 | 136 |

**2a** Does the decrease in the amount of drug A appear to be a zero-order or a first-order process?

A. Cannot tell without logarithms
B. First order
C. Zero order

**2b** What is the rate constant k?

**2c** What is the half-life t½?

3. A solution of a drug was freshly prepared at a concentration of 280 mg/mL. After 30

days at 25°C, the drug concentration in the solution was 84 mg/mL.

**3a** Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

**3b** Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?